Geometric Adaptations of PDE-G-CNNs

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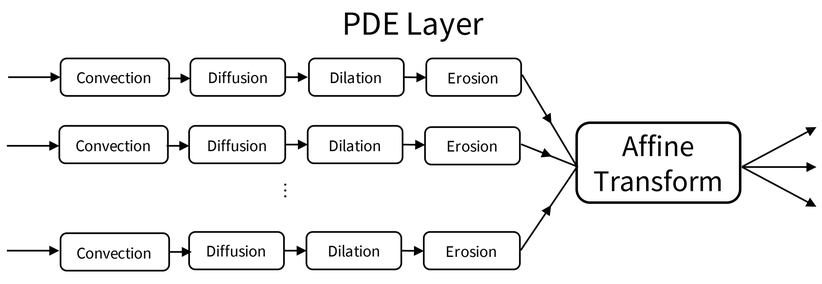
What are PDE-G-CNNs?

In a **PDE-based group equivariant convolution neural network (PDE-G-CNN)** the usual components that make up **CNNs**, that being convolutions, max pooling, and activation functions, are replaced by solvers for **geometrically interpretable partial differential equations (PDEs)**. The parameters that describe the PDEs are learned during training. Moreover, just like **G-CNNs**, PDE-G-CNNs are **group equivariant**. This means, for example, that when an input image is roto-translated one can be **certain** that the output of the network roto-translated accordingly.

	CNN	G-CNN	PDE-G-CNN
Equivariant[1]	✗	✓	✓
Interpretable	✗	✗	✓

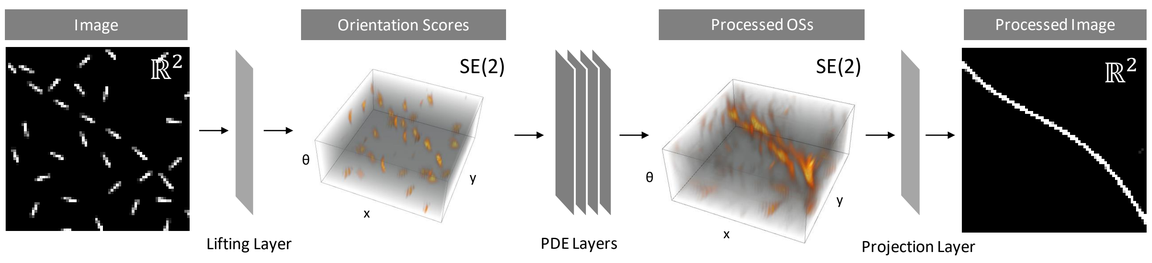
Specifically, the PDEs are **convection, diffusion, dilation, and erosion**. These PDEs respectively correspond to shifting, smoothing, max pooling, and min pooling. They are solved by resampling, linear convolutions, and morphological convolutions.



Name	Convection	Diffusion	Dilation	Erosion
Solver	Resampling	Linear Conv	Morphological Conv	Morphological Conv
PDE	$\frac{\partial W}{\partial t} = -v \cdot \nabla W$	$\frac{\partial W}{\partial t} = \frac{1}{2}\Delta W$	$\frac{\partial W}{\partial t} = \frac{1}{2}\lvert\lvert\nabla W\rvert\rvert^2$	$\frac{\partial W}{\partial t} = -\frac{1}{2}\lvert\lvert\nabla W\rvert\rvert^2$

However, in stark contrast to classical CNNs the feature maps of PDE-G-CNNs **do not live on** $\mathbb{R}^2$. Instead the feature maps are defined on $SE(2)$**: the group of rigid body motions of the plane**. Consequently, the PDEs are in actuality defined on $SE(2)$. This change in architecture is necessary to achieve roto-translation equivariance, while not sacrificing expressivity. We represent points in $SE(2)$ by $(x, y, \theta) \in \mathbb{R}^2 \times [0, 2\pi)$ where (x, y) represent the translation and θ the rotation.

Given that the input and output of the network are images on $\mathbb{R}^2$, but our feature maps are on $SE(2)$, we need a way to translate between the two. The process of going (in an equivariant manner) from $\mathbb{R}^2$ to $SE(2)$ is called **lifting**, and vice versa **projecting**.

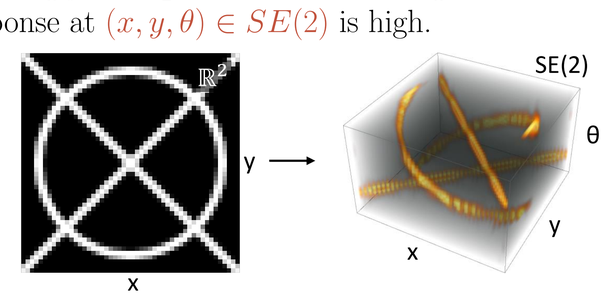


Lifting, Orientations Scores & Cake Wavelets

There is one very natural way to **lift** images using **orientation score transforms**. Intuitively, the orientation score transform $\mathcal{W}$ takes a **kernel** ψ, roto-translates it across the input image f, and "saves" the response as a function on $SE(2)$.

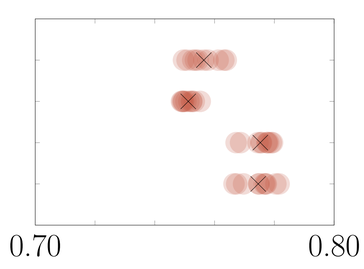
$$(\mathcal{W}_\psi f)(x, y, \theta) := \int_{\mathbb{R}^2} \psi(R_\theta^{-1}(\boldsymbol{y} - [\tbinom{x}{y}]))\, f(\boldsymbol{y})\, d\boldsymbol{y},$$

If an appropriate kernel is chosen to perform the lifting, such as **cake wavelets**[2], the lifted image becomes exceedingly **interpretable**: if the image contains a local orientation θ at position x, y the response at $(x, y, \theta) \in SE(2)$ is high.



In the original PDE-G-CNN paper[3] lifting was performed with **learned kernels**. The interpretability of PDE-G-CNNs would improve drastically if we used the cake wavelets, but by doing this we might lose some performance. Experiments on a vessel segmentation dataset (DCA1) tell us the following: **lifting with cake wavelets increases interpretability while not sacrificing performance**.

Type	Lifting	Params.	Dice Coeff.
CNN	-	25662	0.756
G-CNN	Trained	24632	0.751
PDE-G-CNN	Trained	2560	0.775
PDE-G-CNN	Fixed	2536	0.775



[1] Technically CNNs are translation equivariant, but we are discussing $SE(2)$ equivariance.
[2] An online visualization of cake wavelets can be found at `www.bellaard.com/apps/Cake Wavelets/`
[3] `https://link.springer.com/article/10.1007/s10851-022-01114-x`
[4] The metrics on $SE(2)$ we consider are left-invariant and can therefor completely be determined by a single 3×3 matrix.
[5] The actual distance d is not visualized here. Instead we see an approximation of d called the logarithmic distance approximation ρ_c.
[6] In the paper one more geometric adaptation is considered.

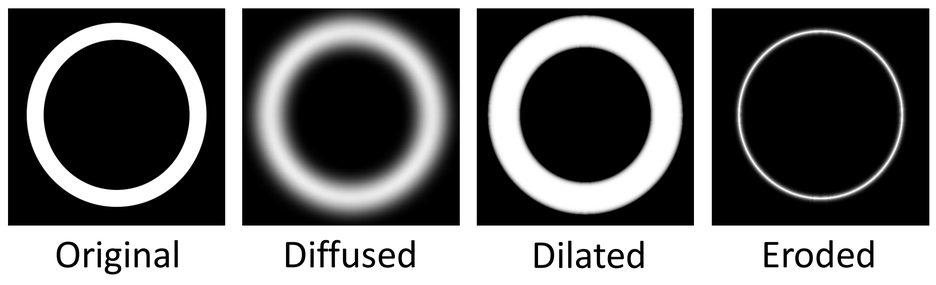
Diffusion, Dilation, Erosion & Convolutions

Diffusion with initial condition $W|_{t=0} = U$ in $\mathbb{R}^2$ can be solved using a **linear convolution** $*$ with a **kernel** k_t. Similarly, **dilation & erosion** in $\mathbb{R}^2$ are solved using a **morphological convolution** $\Box$ with a different kernel.

Name	PDE	Solution	Kernel
Diffusion	$\frac{\partial W}{\partial t} = \frac{1}{2}\Delta W$	$W = k_t * U$	$k_t(\boldsymbol{x}) \propto \exp(-\frac{1}{2t}\lvert\lvert\boldsymbol{x}\rvert\rvert^2)$
Dilation	$\frac{\partial W}{\partial t} = +\frac{1}{2}\lvert\lvert\nabla W\rvert\rvert^2$	$W = -(k_t \Box -U)$	$k_t(\boldsymbol{x}) = \frac{1}{2t}\lvert\lvert\boldsymbol{x}\rvert\rvert^2$
Erosion	$\frac{\partial W}{\partial t} = -\frac{1}{2}\lvert\lvert\nabla W\rvert\rvert^2$	$W = k_t \Box U$	$k_t(\boldsymbol{x}) = \frac{1}{2t}\lvert\lvert\boldsymbol{x}\rvert\rvert^2$

$$(k * f)(\boldsymbol{x}) = \int_{\boldsymbol{y}\in\mathbb{R}^2} k(\boldsymbol{x} - \boldsymbol{y}) \cdot f(\boldsymbol{y})\, d\boldsymbol{y}$$

$$(k \Box f)(\boldsymbol{x}) = \inf_{\boldsymbol{y}\in\mathbb{R}^2} k(\boldsymbol{x} - \boldsymbol{y}) + f(\boldsymbol{y})$$



However, remember that in a PDE-G-CNN these PDEs take place on $SE(2)$. Luckily the solution method generalizes straightforwardly to $SE(2)$: we replace the convolutions with **group convolutions** and replace $\lvert\lvert\boldsymbol{x}\rvert\rvert = d(\boldsymbol{0}, \boldsymbol{x})$ in the kernels k_t with the **Riemannian distance** $d(e, g)$ to the identity element e on $SE(2)$.

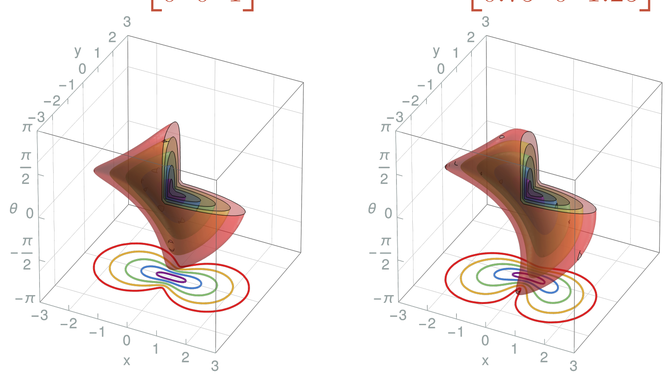
(Non)Diagonal Metrics

But to turn $SE(2)$ into a **Riemannian manifold** we need to endow it with a **metric** $\mathcal{G}$. The group $SE(2)$ is a 3-dimensional manifold, and to fully describe the metric we need a single 3×3 **symmetric positive definite** (SPD) matrix[4]. This can be compared with the definition of an **inner product space**.

The effect of diffusion, dilation and erosion are determined completely by the metric and thus, in turn, determined completely by this matrix. In fact, the **trainable parameters** in a PDE-G-CNN are exactly these matrices. In the original PDE-G-CNN paper only **diagonal** matrices were considered. Allowing for the more general **non-diagonal** matrices might improve performance of the PDE-G-CNNs as the effect of the PDEs is changed considerably.

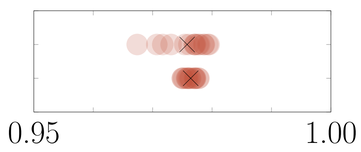
Given that the diffusion, dilation, and erosion kernels k_t are solely functions of the distance d, we can get an idea of how the metric affects the PDEs by visualizing the distance d. We can visualize the distance d by plotting **Riemannian spheres**: i.e. isocontours of the distance d to the identity e[5]. This is analogous to the standard spheres in $\mathbb{R}^3$, that being isocontours of the distance to the origin.

$$\mathcal{G} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \mathcal{G} = \begin{bmatrix} 1.25 & 0 & 0.75 \\ 0 & 16 & 0 \\ 0.75 & 0 & 1.25 \end{bmatrix}$$



Experiments on a synthetic line completion dataset (Lines) tell us the following: **Allowing for non-diagonal metrics has no significant impact on the average performance.**

Metric	Params.	Dice Coeff.
Diagonal	6018	0.9758
Nondiagonal	7458	0.9764



Conclusion

We have investigated two[6] geometric adaptations of PDE-G-CNNs.

- We showed on the DCA1 dataset that one does not need to lift with learned kernels in PDE-G-CNNs. A fixed lifting layer using cake wavelets gives identical performance, while considerably increasing the geometric interpretation of PDE-G-CNNs.
- We showed on the Lines dataset that using nondiagonal metrics gives no significant increase in performance. We hypothesize that the effect of a nondiagonal metric can be (sufficiently) emulated by a diagonal metric combined with convection.

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